

MATHEMATICS PRACTICE EXAMINATION

Subject: Trigonometry • Class: 10

Name: _____ Roll No: _____ Max. Marks: 50

Q.1 If $\cot\theta + \cos\theta = p$ and $\cot\theta - \cos\theta = q$, prove that $p^2 - q^2 = 4\sqrt{pq}$.

Q.2 If θ is an acute angle and $\tan\theta + \cot\theta = 2$, then the value of $\sin^3\theta + \cos^3\theta$ is:

(A) 1

(B) $1/2$

(C) $\sqrt{2}/2$

(D) $\sqrt{2}$

Q.3 Prove the following identity: $2(\sin^6\theta + \cos^6\theta) - 3(\sin^4\theta + \cos^4\theta) + 1 = 0$

Q.4 Find the value of x for which:

$$\frac{3\sin 30^\circ - 4\sin^3 30^\circ}{2\sin^2 50^\circ + 2\cos^2 50^\circ}$$

Q.5 Show that: $\frac{1 + \tan A}{2 \sin A} + \frac{1 + \cot A}{2 \cos A} = \operatorname{cosec} A + \sec A$

Q.6 If $\operatorname{cosec}\theta - \cot\theta = \sqrt{2} \cot\theta$, then prove that $\operatorname{cosec}\theta + \cot\theta = \sqrt{2} \operatorname{cosec}\theta$

Q.7 Prove that: $\sin^2\theta \cdot \tan\theta + \cos^2\theta \cdot \cot\theta + 2 \sin\theta \cos\theta = \tan\theta + \cot\theta$

Q.8 If $\sin\theta + \cos\theta = \sqrt{3}$, then prove that: $\tan\theta + \cot\theta = 1$

OR

Prove that: $(\sin A + \sec A)^2 + (\cos A + \operatorname{cosec} A)^2 = (1 + \sec A \cdot \operatorname{cosec} A)^2$

Q.9 If $\sin A + \sin^2 A = 1$, find the value of: $\cos^2 A + \cos^4 A$. Also, using the above, prove that $\tan^2 A \cdot \sec^2 A = 1$.

OR

Prove that: $\frac{1 + \operatorname{cosec} \theta}{\operatorname{cosec} \theta} = \frac{\cos^2 \theta}{1 - \sin \theta}$

Q.10 Prove that: $\frac{1}{\sec x - \tan x} - \frac{1}{\cos x} = \frac{1}{\cos x} - \frac{1}{\sec x - \tan x}$

Q.11 Prove that: $\frac{\tan \theta}{1 + \tan^2 \theta} + \frac{\cot \theta}{1 + \cot^2 \theta} = 2 \sin \theta \cos \theta$

Q.12 Prove that: $\sin^8 \theta - \cos^8 \theta = (1 - 2\cos^2 \theta)(1 - 2\sin^2 \theta)$

Q.13 Prove that: $\frac{\tan^2 A}{\tan^2 A - 1} + \frac{\operatorname{cosec}^2 A}{\sec^2 A - \operatorname{cosec}^2 A} = \frac{1}{1 - 2\cos^2 A}$

Q.14 Prove that $b^2 x^2 - a^2 y^2 = a^2 b^2$ if:

(i) $x = a \sec \theta$, $y = b \tan \theta$

(ii) $x = a \operatorname{cosec} \theta$, $y = b \cot \theta$

Q.15 Prove that: $(\cot A + \sec B)^2 - (\tan B - \operatorname{cosec} A)^2 = 2(\cot A \cdot \sec B + \tan B \cdot \operatorname{cosec} A)$

Q.16 Prove that: $\tan^2 A \cdot \sec^2 B - \sec^2 A \cdot \tan^2 B = \tan^2 A - \tan^2 B$

Q.17 Prove that: $(\tan A - \tan B)^2 + (1 + \tan A \cdot \tan B)^2 = \sec^2 A \cdot \sec^2 B$

Q.18 Prove that: $(1 + \tan x + \sec x)^2 = 2(1 + \sec x)(\sec x + \tan x)$

Q.19 It is given that: $\sin(A - B) = \sin A \cdot \cos B - \cos A \cdot \sin B$. Use it to find the value of: $\sin 15^\circ$

Q.20 If $\sin A = y$, then express $\cos A$ and $\tan A$ in terms of y .

Q.21 Eliminate θ between the equations: $x = a \sec^3 \theta$ and $y = b \tan^3 \theta$.

Q.22 Show that: $\sin^6 A + 3 \sin^2 A \cdot \cos^4 A = 1 - \cos^6 A$.

Q.23 Prove that: $\frac{1 + \sec \theta - \tan \theta}{1 + \sec \theta + \tan \theta} = \frac{\cos \theta}{1 + \sin \theta}$

Q.24 If $\sin \theta + \cos \theta = \sqrt{3}$, then find the value of: $\sin \theta \cdot \cos \theta$.

Q.25 If $\sin \theta - \cos \theta = 0$, then find the value of: $(\sin^4 \theta + \cos^4 \theta)$.

Q.26 Prove that: $\frac{\operatorname{cosec} A - \cot A}{\operatorname{cosec} A + \cot A} + \frac{\operatorname{cosec} A + \cot A}{\operatorname{cosec} A - \cot A} = 2(2 \operatorname{cosec}^2 A - 1)$
 $= 2 \left(\frac{1 + \cos^2 A}{1 - \cos^2 A} \right)$

Q.27 Prove that: $\frac{\sec^3 \theta}{\sec^2 \theta - 1} + \frac{\operatorname{cosec}^3 \theta}{\operatorname{cosec}^2 \theta - 1} = \sec \theta \cdot \operatorname{cosec} \theta (\sec \theta + \operatorname{cosec} \theta)$

Q.28 In $\triangle ABC$, $\angle B = 90^\circ$, $AB = 8$ cm and $AC = 10$ cm. If CD bisects AB and $\angle DCB = \theta$, then find:

(i) $\sin \theta$ (ii) $\operatorname{cosec} \theta$ (iii) $\tan \theta$ (iv) $\cos^2 \theta - \sin^2 \theta$